# Exploring the Use of a Large Language Model in Simulation Debriefing: An Observational Simulation-Based Pilot Study

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**Introduction:** Facilitating debriefings in simulation is a complex task with high task load. The increasing availability of generative artificial intelligence (AI) offers an opportunity to support facilitators. We explored simulation facilitation and debriefing strategies using a large language model (LLM) to decrease facilitators' task load and allow for a more comprehensive debrief.

**Methods:** This prospective, observational, simulation-based pilot study was conducted at Yale University School of Medicine. For each simulation, a debriefing script was generated by passing a real-time transcription of the simulation case as input to the GPT-4o LLM. Thereafter, facilitators and learners completed surveys and task workload assessments. The primary outcome was the task workload as measured by the NASA-TLX scale. The secondary outcome was the perception of the AI technologies in the simulation, measured with survey-based questions.

**Results:** This study involved four facilitators and 25 learners, with all data being self-reported. All showed strong enthusiasm for AI integration, with mean Likert scores of 4.75/5 and 4.0/5, respectively. NASA-TLX scores revealed moderate to high mental demand for facilitators (M = .8/21; SD = 6.4) and learners (M = 9.9/21; SD = 4.5). AI was perceived to help maintain focus (M = 4.8/5), support learning objectives (M = 4.2/5), and minimize distractions for both facilitators (M = 4.6/5) and teams (M = 4.5/5).

**Conclusions:** This study highlights LLM integration in aiding debriefing by organizing complex information. Though facilitators reported a considerable task load, findings suggest that LLM can enhance simulation-based debrief quality, while there remains a continuous need for human oversight.

**Key Words:** Simulation, artificial intelligence (AI), debriefing, large language model (LLM)

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Artificial intelligence (AI) has the potential to transform the simulation landscape as it has become more accessible and available to simulation educators and researchers. AI encompasses technologies that simulate human intelligence as it learns from experiences, adapts to new inputs, and executes tasks autonomously.[1] Generative AI focuses on creating new content such as text, images, audio, or video.[1,2] Within this domain, large language models (LLMs), such as language model for dialog applications) and generative pretrained transformer (GPT), process and generate human language, producing contextually relevant and syntactically accurate outputs. These models have impacted healthcare education and simulation practices at a rapid pace.[3–6] As a healthcare simulation community, generative AI offers new opportunities to support learners' diverse learning styles[7] and to improve the simulation training experience both for facilitators and learners. LLMs can be used to monitor teaching quality in the implementation, evaluation, and feedback in medical education programs.[8]

Debriefing and facilitation are essential components of successful simulation training and require a high level of skill and experience.[9] Connecting real-time simulation transcriptions, with multiple speakers and timestamps, to custom, GPT–personalized AI chatbots using the underlying technology of GPT involves prompt engineering tailored to the scenario. These prompts include embedded relevant resources and are integrated through GPT-4o, the most recent version of OpenAI. We hypothesized that this integration could support novice facilitators in their debriefing practice through real-time data organization and presentation. This could also help decrease the facilitator's task load and ensure that all important actions are noticed and discussed.

The goal of this study was to explore the use of generative AI (customGPT with GPT-4o) in pediatric simulations, assess its impact on facilitator task load using the NASA-TLX tool[10] and evaluate perceptions and time allocation during AI-supported debriefs through surveys and time analysis. We hypothesized that customGPT would reduce facilitator task load and be positively received by both facilitators and learners.

## METHODS

### Study Design

This study was institutional review board-approved obtained at Yale University (2000037905) and verbal consent was obtained from all participants. This prospective, observational, simulation-based pilot study took place at Yale University School of Medicine. The study was held on an academic day when pediatric residents were relieved of their clinical duties, allowing inclusion across all postgraduate years (PGYs). Additionally, all curriculum facilitators were eligible for enrollment. As part of the pediatric residents' routine academic curriculum, facilitators and learners provided informed consent before being oriented to the educational intervention and study procedures. Through a heterogeneous rotational design, learners rotated through four stations, each addressing different clinical scenarios or educational objectives. This design allowed all learners to be exposed to several educational objectives as part of their curriculum. Only one station (station 1) included an AI-based intervention, while the others focused on non-AI communication-based simulations. The residents were randomly grouped into small teams (5–6 residents) with one facilitator per group, with one learner engaging in the simulation conversation while the others observed. An embedded standardized actor portrayed the caregiver and followed a semistructured script (see Text, Supplemental Digital Content 1, http://links.lww.com/SIH/B149, which demonstrates Semi-structured Script for Caregiver Actor). Each group rotated through the stations in a sequential manner. This type of design ensured that all learners were exposed to the same interventions but at different times. Each station was allotted a duration of 40 minutes.

## Study Intervention

At the initiation of the study, we obtained baseline and demographic information from both the facilitators and the learners. This included questions about their years of experience with simulation facilitation, frequency of simulation facilitation, experience with AI integration in simulation, and training background in AI.

A mannequin representing an infant with a bruise was used to direct learners' attention to the scenario. The patient simulator used was Laerdal SimNewB (Laerdal Medical, Stavanger, Norway). All facilitators were healthcare professionals who were provided with the case background, standardized caregiver information, patient information, and potential dialog before the simulation (see Text and Table, Supplemental Digital Content 2, http://links.lww.com/SIH/B150, which outlines Custom GPT Prompt). The actors of the caregiver followed a semistructured script, with variations in the questions or statements.

The participants in caregiver and resident roles were positioned centrally as depicted in Figure 1. A Zoom audio device was placed nearby for real-time transcription purposes. The remaining residents in the group observed the simulation, while the facilitator sat between the learner in the resident role and the observers. Although some residents were observers during the facilitation, all actively participated in the debrief. An AI engineer (first author of this study, holding a bachelor's degree in computer science and a master's degree specializing in large language models) at the edge of the room operated the LLM, using real-time transcription from the Zoom audio device on their laptop. Upon completion of the simulation, the debrief was printed for the facilitator and displayed on a big screen, including simulation observers (Fig. 1). To minimize distractions, the facilitator, engineer, LLM, and portable printer were positioned in close proximity.

The simulation scenario selected was of an infant with bruises who presented to the emergency department with his mother (see Text, Supplemental Digital Content 1, http://links.lww.com/SIH/B149, which demonstrates semistructured script for caregiver actor). Each simulation was planned to last about 10 minutes, followed by a 25-minute debrief and a 5-minute survey. A single facilitator conducted the debriefing, supported by an AI-generated debrief script. The script, provided to the facilitator in a 1–2 page printout (see Text and Table, Supplemental Digital Content 3, http://links.lww.com/SIH/B151, which presents AI-generated debrief and evaluation table for facilitators), was also displayed on a large screen (Fig. 1). The facilitator and learners had discretion in referencing the AI-generated debrief script and evaluation checklist. Generative AI was provided with the transcribed debrief to analyze the length of time used for debrief, the coverage of each learning objective, and the adherence to debriefing best practices. The structured debrief was divided into 5 subcomponents (setting the scene, reaction, description, analysis, and summary) according to the Promoting Excellence and Reflective Learning in Simulation framework[11]

After the debrief was completed, facilitators and learners self-reported their experiences with the AI-supported debrief through a survey, which included questions about their perceptions and comfort using a 5-point Likert scale (see Table, Supplemental Digital Content 6, http://links.lww.com/SIH/B154, which illustrates postsurvey) and free-text responses. In addition, facilitators' self-reported task load was assessed using the adjusted NASA Task Load Index (NASA-TLX)[10], a tool designed to measure task load (see Text, Supplemental Digital Content 5, http://links.lww.com/SIH/B153, which illustrates NASA-TLX Survey).

## AI Technology

CustomGPT, OpenAI's customization and prompting feature of GPT-4o, enables users to create custom AI chatbots tailored to specific tasks or specialized knowledge bases without requiring programming expertise. This technology offers flexibility in modifying the knowledge base and adjusting parameters to fine-tune the chatbot's response style. In this study, we leveraged CustomGPT to enhance the educational simulation process by fine-tuning the model using domain-specific content. This included simulation case files, objectives, detailed simulation scenarios, debriefing resources, best practices, case-specific simulation literature (prereading), and a structured evaluation checklist (see Text and Table, Supplemental Digital Content 2, http://links.lww.com/SIH/B150, which outlines custom GPT prompt). These resources were integrated into CustomGPT's database to ensure that the AI could generate highly relevant, content-driven responses in line with the specific simulation context.

The optimization of CustomGPT's performance involved a systematic process of iterative testing and fine-tuning of key GPT parameters, with particular emphasis on the temperature setting, which controls the level of randomness in the AI's output.[12] At each step in generating text, the model predicts the next word by evaluating a range of possible options based on prior context. The temperature setting adjusts how likely the model is to select the most probable word or a less conventional alternative.[13] A temperature of 0 results in highly deterministic outputs, consistently choosing the most likely option, while a temperature of 1 introduces greater randomness, sampling from a broader range of possibilities.[14] For this study, we set the temperature at 0.5 to achieve an optimal balance, ensuring the facilitation scripts generated by the AI were neither rigid

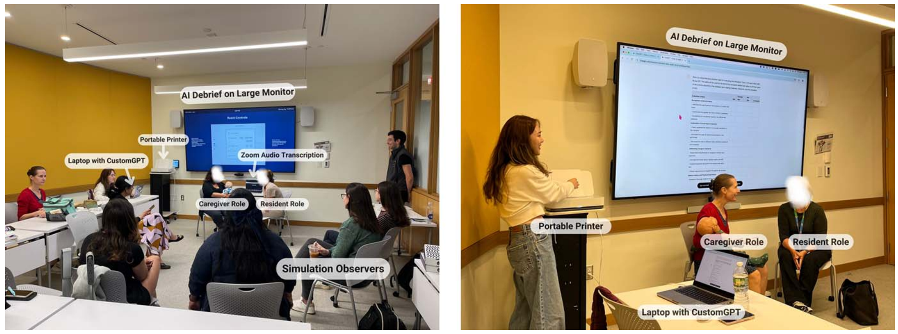


**FIGURE 1.** AI integration simulation setup. Simulation setup using custom LLM includes distinct roles for caregivers, residents, facilitators, and simulation observers. The setup also features Zoom audio transcription, a laptop equipped with custom GPT, a portable printer, and an AI-supported debrief displayed on a large monitor.

reproductions of preloaded content nor overly divergent.[12] This approach provided adaptable outputs that effectively supported the facilitators' task load and available resources.

## Outcomes

The primary outcome was the facilitators' and learners' task workload. This was assessed through the adapted NASA-TLX, a questionnaire focusing on task load across multiple domains, including mental and physical demand, temporal demand, performance, effort, frustration, and anxiety (see Table, Supplemental Digital Content 4, http://links.lww.com/SIH/B152, which lists NASA-TLX task load types). The secondary outcome was the subjective perception of incorporating AI technologies into the simulation, measured with survey-based questions using a 5-point Likert scale (see Table, Supplemental Digital Content 6, http://links.lww.com/SIH/B154, which illustrates postsurvey) The survey was analyzed separately for facilitators and learners, respectively.

## Statistical Analysis

The survey and NASA-TLX were analyzed using descriptive statistics (ie, frequencies, means, and standard deviations) for key demographics and outcomes. As this was a pilot study, there was no a priori power analysis and sample size calculation; a convenience sample of 4 facilitators and 25 learners was used. Figures were made with Prism v. 10 (GraphPad, Boston, MA).

# RESULTS

## Study Population

We enrolled four facilitators and 25 learners, all experienced in traditional simulation but novices to AI-supported simulations (Tables 1, 2). None of the facilitators had previously used AI technologies in scenario design, facilitation, or debriefing, nor had they received formal AI training. A similar trend was observed among learners, with 1 having experience in AI for simulation facilitation and 2 having experience in AI for simulation debriefing. Despite this, facilitators showed strong enthusiasm for AI integration (mean score = 4.75/5), and learners also showed interest (mean score = 4.0/5).

## NASA-TLX Scores

Facilitator workload was evaluated using the NASA Task Load Index (NASA-TLX). The total mean NASA-TLX score for facilitators was 40.8 (SD = 4.6) (out of a max of 147). Effort scored the highest at 10.8 (SD = 2.5), followed by mental demand at 7.8 (SD = 6.4), compared with other task load parameters. Physical and temporal demands were reported as relatively low, at 4.0 (SD = 4.8) and 2.8 (SD = 2.1), respectively (Table 2). Learners also experienced higher mental demand, with a mean score of 9.9 (SD = 4.5), and similar levels of effort (9.4, SD 4.7). They faced higher cognitive demands than facilitators. Notably, learners demonstrated average performance scores (M = 7.8, SD = 3.7; Table 3), while facilitators scored (M = 5.3, SD = 4.2), which is below average. However, as the performance scale is reversed (see Text, Supplemental Digital Content 5, http://links.lww.com/SIH/B152, which illustrates NASA-TLX Survey), a lower score indicates better performance. This suggests that higher cognitive load does not necessarily correlate with lower performance scores.

## Perception Survey Report

The facilitators' and learners' perceptions of AI technology during the debriefing session are summarized in Figure 1. The data highlight the nondisruptive nature of AI integration into the debriefing process. Both groups generally agreed that AI did not disrupt the transition from the scenario to the debrief (mean score = 4.8/5) and did not interfere with the flow of the debrief (mean score = 4.5/5). Specifically, the AI system was rated highly for not distracting the facilitator (mean score = 4.6/5) and for not distracting the team during the debrief (mean score = 4.5/5).

# DISCUSSION

We explored the use of a LLM to support the debrief during pediatric simulations. The findings demonstrated that LLM integration not only functioned well from a technical perspective, with a smooth connection between real-time transcription and debrief generation, but also operated noninvasively, without causing distractions for either facilitators or learners during the simulation. Both facilitators and learners reported that the AI-assisted debriefs enhanced the organization of complex information, ensuring that key learning objectives were comprehensively addressed. These findings align with the moderately good performance scores, particularly for facilitators and reflect the potential of AI in facilitating a balanced debriefing process that enhances educational outcomes without introducing distractions or unnecessary complexity.

**TABLE 1.** Demographics of Facilitators

| Demographic | n = 4 (%) |
|---|---|
| How many years have you been a simulation facilitator? | |
| <1 yr | 0 (0) |
| 1–5 yr | 2 (50) |
| 6–10 yr | 1 (25) |
| 10–15 yr | 1 (25) |
| >15 yr | 0 (0) |
| How many simulations would you estimate do you facilitate per year | |
| never | 0 (0) |
| 1–5 times | 2 (50) |
| 6–10 times | 1 (25) |
| 10–15 times | 1 (25) |
| >15 times | 0 (0) |
| How many times would you estimate that you used AI technology for scenario design? | |
| never | 4 (100) |
| 1–5 times | 0 (0) |
| 6–10 times | 0 (0) |
| 10–15 times | 0 (0) |
| >15 times | 0 (0) |
| How many times would you estimate that you used AI technology for simulation facilitation? | |
| never | 4 (100) |
| 1–5 times | 0 (0) |
| 6–10 times | 0 (0) |
| 10–15 times | 0 (0) |
| >15 times | 0 (0) |
| How many times would you estimate that you used AI technology for simulation debrief? | |
| never | 4 (100) |
| 1–5 times | 0 (0) |
| 6–10 times | 0 (0) |
| 10–15 times | 0 (0) |
| >15 times | 0 (0) |
| Have you had any training in AI technologies? | |
| Yes | 0 (0) |
| No | 4 (100) |
| Enthusiasm (1–5), mean (SD) | 4.75 (0.5) |

In line with the findings of Holderried et al (2024),[15] our study showed that learners highly valued the AI's ability to provide immediate, structured feedback, which significantly enhanced the organization and flow of discussions. In their study, GPT-4 successfully simulated patient interactions with over 99% medically plausible responses and strong agreement with human raters (Cohen $\kappa = 0.832$). This ability to deliver reliable, real-time feedback closely mirrors our findings, where the AI-supported debriefing sessions became more structured and efficient. These parallels underscore the potential of LLMs to enhance simulation-based communication training by delivering timely, objective feedback that improves learner engagement and learning outcomes. In addition, LLM integration allowed facilitators to navigate sensitive topics, such as child abuse scenarios.

Furthermore, we assessed the facilitators' task load during LLM supported debriefs. Despite AI's intention to reduce task load, facilitators reported medium mental demand, indicating that AI integration may not fully alleviate the cognitive challenges of complex debriefing. NASA-TLX scores revealed that facilitators experienced mental demand during the AI-supported debriefs, suggesting that their role remained cognitively taxing despite AI assistance (Fig. 2). It is important to note that this study is observational and lacks a control group. However, to better contextualize the NASA-TLX scores, it may be helpful to compare them to other simulation studies examining debriefer task load with a debriefing tool, namely cognitive aid, such as Meguerdichian (2022).[16] Notably, all types of task load reported in our study are lower than that of the control group in the Meguerdichian (2022)[16] study (Table 3), suggesting that while AI integration does not eliminate cognitive demand, it may contribute to a more structured debriefing process. Larger, more rigorous studies are necessary to validate these initial findings.

These results suggest that while LLM can ease certain aspects of the facilitation process, managing an AI-supported debrief remains a cognitively demanding task. That is, while the integration of AI-based systems in medical simulations is effective in automating certain processes, it still requires human oversight to ensure clinical accuracy and relevance.[17] Nonetheless, LLM significantly helped facilitators lead the debrief more effectively and reduced the need for manual tracking of discussion points (Fig. 3), potentially easing the task load associated with organizing real-time information. Most importantly, many learners noted that despite the additional cognitive demand, the integration of AI provided valuable, constructive feedback. One learner highlighted that the AI offered “a checklist and nonpartial, unbiased feedback—both positive and constructive,” which significantly enhanced the debriefing process (see Table, Supplemental Digital Content 7, http://links.lww.com/SIH/B155, which presents survey free-text response). Another learner appreciated how AI complemented the attending facilitator, stating that it helped “debrief and provide constructive criticism,” suggesting the AI's role in creating a more

**TABLE 2.** Demographics of Learners

| Demographic | N = 25 (%) |
|---|---|
| Training | |
| PGY2 | 12 (48) |
| PGY3 | 13 (52) |
| No. simulations you have participated in? | |
| 1–5 times | 2 (8) |
| 6–10 times | 8 (32) |
| 10–15 times | 7 (28) |
| >15 times | 8 (32) |
| Have you participated in a simulation where AI technologies were used for the simulation facilitation? | |
| No | 24 (96) |
| Yes | 1 (4) |
| Have you participated in a simulation where AI technologies were used for simulation debrief? | |
| No | 23 (92) |
| Yes | 2 (8) |
| Enthusiasm (1–5), mean (SD) | 4.0 (1.0) |

**TABLE 3.** NASA-TLX Scores

| Task Load | Mean NASA TLX Domain Score (SD) | Scores Ranged From 0 to 100 |
|---|---|---|
| **Facilitators** | **n = 4** | |
| Mental demand | 7.8 (6.4) | 37.1 |
| Physical demand | 4.0 (4.8) | 19.0 |
| Temporal demand | 2.8 (2.1) | 13.3 |
| Performance | 5.3 (4.2) | 25.2 |
| Effort | 10.8 (2.5) | 51.4 |
| Frustration | 4.3 (4.7) | 20.5 |
| Anxiety | 5.8 (4.6) | 27.6 |
| average | 5.8 (4.40) | 27.73 |
| Total | 40.8 (4.6) | 194.1 |
| **Learners** | **N = 25** | |
| Mental demand | 9.9 (4.5) | 47.1 |
| Physical demand | 2.1 (2.3) | 10.0 |
| Temporal demand | 5.5 (4.1) | 26.2 |
| Performance | 7.8 (3.7) | 37.1 |
| Effort | 9.4 (4.7) | 44.8 |
| Frustration | 3.0 (3.0) | 14.3 |
| Anxiety | 7.5 (5.8) | 35.7 |
| Average | 6.5 (4.15) | 30.74 |
| Total | 45.2 (4.9) | 215.2 |

objective and structured environment despite the ongoing cognitive challenges (see Table, Supplemental Digital Content 7, http://links.lww.com/SIH/B155, which presents survey free-text response). This feedback aligns with prior studies emphasizing the need for comprehensive LLM evaluation frameworks that go beyond accuracy to include considerations of bias, fairness, robustness, efficiency, calibration, and toxicity.[18,19]

Lastly, we evaluated both facilitators' and learners' perceptions of AI-supported debriefs. Post-survey results demonstrated high levels of enthusiasm among learners. This remained consistent with both facilitators' and learners' prior enthusiasm for the AI integration simulation experience (Tables 1, 2). They both acknowledged that the AI's input effectively addressed all critical discussion points, thereby alleviating the task load on facilitators, who otherwise would have had to manually track each point. Moreover, learners expressed excitement about integrating LLMs into future simulations and applying them in their own practice settings, emphasizing the AI's broader relevance and impact on simulation-based education (see Table, Supplemental Digital Content 7, http://links.lww.com/SIH/B155, which presents survey free-text response). Similarly, GPT-driven chatbots enhanced engagement and delivered plausible responses during history-taking exercises, a crucial aspect of medical communication training.[20]

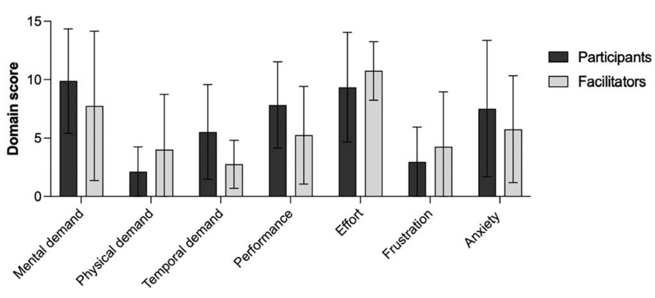


**FIGURE 2.** NASA TLX scores. Means are represented with error bars representing standard deviations.

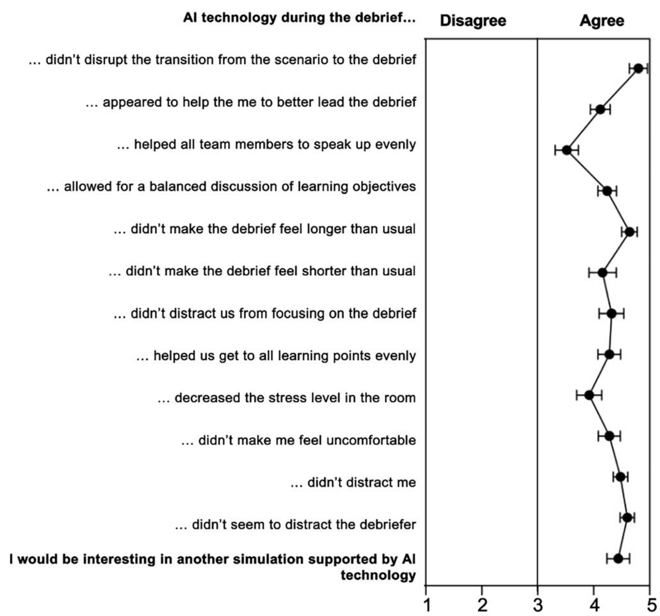


**FIGURE 3.** Postsurvey results.

Overall, LLM integration improved debriefing by helping facilitators track key learning objectives and manage information in real-time. It kept discussions organized, objective, and constructive. Though facilitators experienced moderate to high cognitive load, both they and learners appreciated how the AI streamlined discussions and covered important topics, showing its potential to transform simulation-based education.

## Limitations

While this study highlights a new way to incorporate LLM into our simulation practice, several limitations should be acknowledged to better interpret the findings. First, this was a pilot study conducted at a single-site with a relatively small sample size. While this is appropriate for a pilot study, it does restrict the generalizability of the findings to broader medical education settings. The observational design prevents establishing causality between AI-supported debriefing and observed outcomes and the self-reported Likert scale may not fully capture participant and facilitator perceptions. A future study using rigorous qualitative methods could provide deeper insights, while a comparison with non-AI groups would help clarify outcomes. Furthermore, a comparison study with non-AI groups will be needed to better investigate the outcomes. As an exploratory pilot study without a priori power analysis, the results should be interpreted with caution and cannot be considered definitive. A randomized controlled trial would address both generalizability and it would help establish causality. Additionally, given the cross-sectional nature of the study, it was not possible to evaluate whether AI directly improved the organization and flow of discussions over time. Lastly, our study is at risk for selection bias given the self-reported measures such as surveys, where participant enthusiasm for the novel AI technology could have affected responses.

## CONCLUSIONS

This study, piloting generative AI-supported pediatric simulation, lays the groundwork for future research into AI's role in medical education. The results highlight the potential of generative AI to enhance debriefing organization, though further research is needed to optimize its ability to reduce facilitator workload. Future directions should include randomized controlled trials with larger, more diverse populations and the development of tailored AI interventions. Ultimately, incorporating ongoing feedback from facilitators and learners will be essential in refining AI's role and expanding its utility across various simulation scenarios in medical education.

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